Bangladesh University of Engineering and Technology (BUET)

Assignment

DFT simulation

Course Code: EEE 6512

Course Title: Nanoscale Device Modeling and Simulation Techniques

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Instructions

Use DFT to find out the band structure of Silicon and its effective masses.

You may use any DFT software of your choice (eg. Materials Studio, Quantum Espresso, Gaussian, ADF, Schrodinger etc.)

List the procedure step-by-step, parameters and outputs.

Procedure:

- First Simulate brand structure of Si using Material Studio.
- Get the conduction band and valance band data from the graph. As band gap of Si is 1.12eV we need to select proper band from the whole diagram. Upper valley of valence band and lower valley of conduction band was chosen.
- After getting data from band diagram effective mass for electron and hole was calculated using MATLAB software.

Result:

Band Gap was found 1.16eV from E-K diagram. Effective mass of electron and hole was obtained 4.1594e-31 and 7.5835e-31 respectively which is very similar to original data found in website.

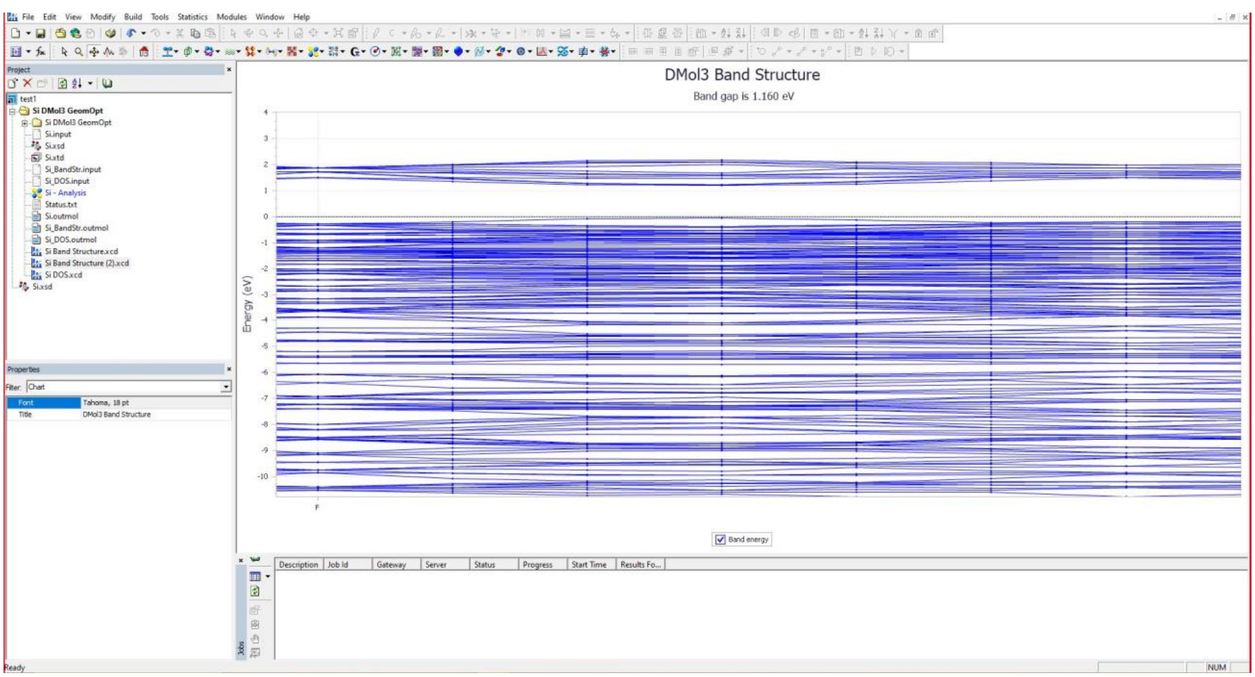

Fig-1: Band structure of Silicon

Appendix:

```
clear all;
close all;
clc;

%valance band data
y1=1.6e-19*[0
-0.059021491
-0.195078408
-0.281039168
-0.198343774
-0.077334752
-0.043919173
-0.128873112
-0.230616474
-0.200983278
-0.199296172
-0.218616254
-0.227514377
-0.073062564
0
];


%Conduction band data
y2=1.6e-19*[1.159803583
1.211478
1.354419397
1.508082079
1.354120072
1.219641415
1.19789952
1.25632236
1.380351846
1.509361014
1.420924018
1.50612286
1.362174642
1.217246814
1.159803583
];
```

```
%k Axis data
x=[0
0.06779172
0.135579372
0.203371092
0.273870516
0.344367147
0.414866571
0.485367113
0.555866538
0.626363169
0.696862593
0.772646035
0.848431296
0.924216558
1
]*(2*pi)./(3.17e-10);

figure(1);
plot(x,y1,'r');
hold on;
plot(x,y2,'g');

h_cut= 6.626e-34/(2*pi);


%Effective mass for electron
der1_con= diff(y2)./diff(x);
der2_con = diff(der1_con)./diff(x(1:14));
inv_der_con = 1./der2_con;
eff_mass_con = (h_cut*h_cut).*inv_der_con;



%Effective mass for hole
der1_val= diff(y1)./diff(x);
der2_val = diff(der1_val)./diff(x(1:14));
inv_der_val = 1./der2_val;
eff_mass_val = (h_cut*h_cut).*inv_der_val;

eff_mass_electron = min(abs(eff_mass_con))
eff_mass_hole = min(abs(eff_mass_val))
```